

The Core Lemma reduces to an all-stay inequality: the min- D Key Lemma

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Abstract

We reduce the Core Lemma (the minimal-degree lower bound $\text{mincost}(\lambda) \geq n(\lambda)$ for $\chi^\lambda(\Omega)$) to a single per-walk statement, the *Key Lemma*: every feasible closed walk along the staircase word costs at least $\min_t D(U_t)$, the smallest all-stay cost among its visited tableaux. Combined with the already proven all-stay inequality $D(T) \geq n(\lambda)$ (valid for *every* tableau), the Key Lemma yields the Core Lemma. We prove a pairing lemma (sorting and de-sorting swaps are equinumerous in any closed walk), deduce that the two natural cost conventions coincide and that $\text{cost} = \#\{\text{descending encounters}\}$, prove the reduction, and verify the Key Lemma exactly for $n \leq 7$ (exhaustively for $n \leq 6$). We also split the Key Lemma at the argmin into two verified claims and rule out the natural closed-form potentials.

1 Setup

$\lambda \vdash n$, rows/columns 0-indexed, $\text{cont}(r, c) = c - r$, and $\text{cont}_T(v)$ the content of the cell holding v in $T \in \text{SYT}(\lambda)$. Axial distance $d_T(i) = \text{cont}_T(i+1) - \text{cont}_T(i)$ and descent indicator $\delta_T(i) = [d_T(i) < 0]$ (adjacent values never share a diagonal, so $d_T(i) \neq 0$). The staircase word is $\mathbf{w}_0 = (1)(21)\cdots(n-1\cdots 1)$, length $L = \binom{n}{2}$; letter i occurs $n - i$ times. A closed walk is $U_0 \rightarrow \cdots \rightarrow U_L = U_0$, one step per letter i_k , each step a STAY ($U_k = U_{k-1}$) or SWAP ($U_k = s_{i_k} U_{k-1}$, allowed iff $|d_{U_{k-1}}(i_k)| \geq 2$). It is *feasible* iff no before-state has $d_{U_{k-1}}(i_k) = -1$.

Cost (after-state). The $q \rightarrow 0$ orders of the Hoefsmit matrix $M = T_i + 1$ make the cost of a step the order of the entry in column U_k , i.e. $\text{cost}_k = \delta_{U_k}(i_k)$. Thus: STAY at $d > 0$ costs 0; STAY at $d < 0$ costs 1; de-sorting SWAP (from $d \geq 2$) costs 1; sorting SWAP (from $d \leq -2$) costs 0. The all-stay cost of a fixed tableau is $D(T) = \sum_k \delta_T(i_k) = \sum_i (n - i) \delta_T(i)$.

All-stay lemma (proven; cited). $D(T) \geq n(\lambda)$ for every $T \in \text{SYT}(\lambda)$, equality at T_{rs} (column-tower argument, Lemma 1 of the companion all-stay note). No feasibility hypothesis is used.

2 Pairing lemma and cost reformulations

For $U \in \text{SYT}(\lambda)$ let $a_U = (\text{cont}_U(1), \dots, \text{cont}_U(n))$ be the content-sequence and $F(U) = \#\{j < j' : \text{cont}_U(j) > \text{cont}_U(j')\}$ its inversion number.

Lemma 1. A SWAP at comparator i exchanges exactly the entries $a_U(i), a_U(i+1)$ and fixes the rest; hence F changes by $+1$ on a de-sorting swap and -1 on a sorting swap, and is unchanged by a STAY.

Proof. s_i swaps the cells of i and $i+1$, i.e. transposes positions $i, i+1$ of a_U . An adjacent transposition changes the inversion count by ± 1 , with sign $+1$ iff the pair was increasing (the de-sorting case). Verified for all SYT, $|\lambda| \leq 7$. $\square$

Proposition 2. *In every closed walk, $\#\{\text{sorting swaps}\} = \#\{\text{de-sorting swaps}\}$.*

Proof. $F(U_0) = F(U_L)$, so $0 = \sum_k \Delta F_k = \#\text{de-sorting} - \#\text{sorting}$. $\square$

Proposition 3. *For a closed walk,*

$$\text{cost} = \#\{\text{cost-1 stays}\} + \#\{\text{de-sorting swaps}\} = \#\{\text{cost-1 stays}\} + \#\{\text{sorting swaps}\} = \#\{\text{cost-1 stays}\} + \frac{1}{2} \#\{\text{swap}\}$$

Equivalently $\text{cost} = \sum_k \delta_{U_{k-1}}(i_k) = \#\{k : \text{comparator } i_k \text{ is content-descending in } U_{k-1}\}$.

Proof. A cost-1 stay is descending before and after; a de-sorting swap is descending after; a sorting swap is descending before. So the after-state count is (cost-1 stays)+(de-sorting) and the before-state count is (cost-1 stays)+(sorting); these are equal by Proposition 2, and equal their average. $\square$

3 Reduction: Key Lemma $\Rightarrow$ Core Lemma

Conjecture 4 (Key Lemma; verified $n \leq 7$). *Every feasible closed walk satisfies $\text{cost} \geq \min_{0 \leq t \leq L} D(U_t)$.*

Theorem 5 (Core Lemma, conditional). *Assuming Conjecture 4, $\text{mincost}(\lambda) \geq n(\lambda)$ for every λ .*

Proof. For any feasible closed walk W , $\text{cost}(W) \geq \min_t D(U_t)$ by the Key Lemma. Each visited U_t satisfies $D(U_t) \geq n(\lambda)$ by the all-stay lemma (which holds for all SYT), so $\min_t D(U_t) \geq n(\lambda)$ and $\text{cost}(W) \geq n(\lambda)$. Minimise over W . $\square$

4 The split at the argmin

Let t^* minimise $D(U_t)$ and $W = U_{t^*}$. Writing $c_{t^*}(W) = \sum_{j \leq t^*} \delta_W(i_j)$ and $R_{t^*}(W) = \sum_{j > t^*} \delta_W(i_j)$ (so $c_{t^*}(W) + R_{t^*}(W) = D(W)$), and splitting $\text{cost} = \text{cost}_P + \text{cost}_Q$ at t^* :

Claim 6 (prefix). $\text{cost}_P \geq c_{t^*}(W)$.

Claim 7 (suffix). $\text{cost}_Q \geq R_{t^*}(W)$.

Together they give the Key Lemma: $\text{cost} = \text{cost}_P + \text{cost}_Q \geq c_{t^*}(W) + R_{t^*}(W) = D(W) = \min_t D(U_t)$.

Status. Claims 6,7 are verified exhaustively over all feasible closed walks for $n \leq 6$. They are *not* instances of a generic segment lemma: “ D -minimal endpoint of a contiguous-subword segment $\Rightarrow \text{cost} \geq \text{all-stay of that endpoint}$ ” is *false* (counterexamples from $n = 3$). Hence A/B use the globality of W over the whole walk, and presumably feasibility.

Remark 8 (the obstruction). At one swap, D can jump by up to $\pm 3(n-i)$ (it touches comparators $i-1, i, i+1$ with weights $n-i+1, n-i, n-i-1$), while the step cost changes by 0 or 1. So D is not Lipschitz in cost: a single sorting swap may drop D far below the cost paid. This kills every naive potential (§5) and forces the proof to use the global return constraint.

5 Negative results

Each candidate node-potential $\phi_k(T)$ below was tested on all layered edges for all feasible shapes $n \leq 7$ and fails: (1) $\text{cost} \geq D(\text{start})$ is false; (2) the natural $\phi_k = n(\lambda) - R_k(T)$ (boundary correct, stays tight) is violated on every D -decreasing swap; (3) forward-propagation from the natural boundary collapses; (4) any k -independent $\phi_k = c_k(T) + \psi(T)$ creates negative difference-cycles; (5) the symmetric $\phi_k = c_k(T) + \frac{1}{2}F(T)$ is violated on swaps; (6) the running-minimum invariant $\Theta_k = \text{cost}_{[1,k]} + R_k(U_k) \geq \min_{t \leq k} D(U_t)$ fails mid-walk (its endpoint instance is the Key Lemma and is true). A valid integral potential exists per shape (LP dual) but is genuinely k -dependent; the natural closed form is the value function of the two-parameter DP on (tableau, running-min- D), which proves the Key Lemma for each fixed n but not uniformly.

6 Verification

Using the validated Hoefsmit module `compute_p_lambda.py`: the two cost conventions agree and equal $n(\lambda)$ for all feasible shapes $n \leq 7$; the pairing mechanism (Lemma 1) holds for all SYT $|\lambda| \leq 7$; the Key Lemma holds by DP over (state, running-min- D) for $n \leq 7$ and by direct exhaustive enumeration for $n \leq 6$ (zero violations); Claims A,B hold exhaustively for $n \leq 6$. Feasible closed walks are strikingly rare (e.g. $(2, 2, 2)$ admits one; $(4, 2)$ admits 24): feasibility plus closure is very rigid, which is the likely lever for a proof.

The remaining gap

Prove: every feasible closed walk has $\text{cost} \geq \min_t D(U_t)$ (equivalently Claims A,B), using globality of the D -minimal W and the $d = -1$ -avoidance (currently unused). Routes: (i) characterise the rare feasible closed walks directly; (ii) closed-form for the two-parameter DP value function; (iii) straighten/induct on swaps via the sorting/de-sorting pairing (Proposition 2).